The Homology Cliff in Frozen ESM-2 Biosecurity Retrieval: Characterization, Pre-Registered Null Controls, Cross-Family Mechanism, and a Deployable Rescue

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April 12, 2026 (v1.1)

Abstract

Frozen protein language models (PLMs) increasingly feed biosecurity retrieval pipelines. We characterize a systematic failure mode, the *homology cliff*, in frozen ESM-2 [1] cosine k-NN retrieval on a 24,885-protein biosecurity-relevant test set (28.7% positive prior). At scale t30 (150M parameters), reference panel size $R = 1000$, neighbors $k = 25$: close-stratum $F_1 = 0.866$ against distant-stratum $F_1 = 0.120$ (gap +0.745). We report six pre-registered experiments. A 3000-cell main factorial confirms the cliff with all 300 groups passing a pre-committed seed-variance gate. A pre-registered full-pool permutation null (SHA256 `f3864d09...`) produces gap $\in [-0.027, +0.038]$ across 300 groups, every 95% CI including zero: the cliff is not a stratification artifact. The failure is a *precision* collapse: at t30 $R = 1000$ $k = 25$ cosine, distant-stratum positive-prediction precision is $3/44 = 0.068$ and expected calibration error rises $4\times$ from close (0.069) to distant (0.294). Pfam partition of 20 evaluable distant false alarms finds **zero within-family and 20 cross-family** failures, ruling out within-family distant-homology as the mechanism. Four rescue attempts fail at pre-registered H1: Ledoit-Wolf Mahalanobis whitening [2], Fisher-Rao within-class whitening [3], stratified cosine+Mahalanobis cascade, and Mapper-biased panel augmentation. One rescue succeeds: a panel-only linear projection trained via a 50-iteration triplet-surrogate loss [4] wins pooled F_1 in 18 of 18 factorial groups and achieves $F_1 = 0.891$ at t30 $R = 1000$ $k = 25$, improving distant-stratum F_1 by 48% relative. We release 9,360 per-cell bootstrap-CI results, four SHA256-locked pre-registrations, and full code under MIT + CC-BY-4.0.

1 Introduction

Frozen protein language models [1, 5–7] now serve as general-purpose feature extractors for downstream retrieval, classification, and design tasks. A common deployment pattern is cosine k-NN retrieval: given a query protein, retrieve the k nearest panel members in frozen embedding space and vote their labels. For biosecurity applications, where the query set contains potentially novel threats and the panel contains known-toxin exemplars, retrieval performance on proteins distant from any panel member is the safety-critical metric.

We show that cosine k-NN retrieval on frozen ESM-2 embeddings exhibits a systematic *homology cliff*: performance collapses on the distant stratum in a way that grows monotonically with both model scale and panel size. The cliff persists under four independent rescue attempts (post-hoc whitening, per-stratum distance switching, topologically-biased panel augmentation) and under

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cross-family analysis is shown to be an embedding-space geometry artifact, not a missing-homolog artifact. We identify one class of rescue that works: a cheap supervised projection of the embedding space, trained only on the panel, fit in under five CPU seconds.

Contributions:

1. A 3000-cell pre-registered factorial (SHA256 `139f6012...`) characterizing the cliff across (scale, R, k, metric), 10 seeds, with a pre-committed seed-variance gate.
2. A 3000-cell pre-registered full-pool permutation null (SHA256 `f3864d09...`) establishing that the cliff is not a stratification artifact.
3. Mechanistic analysis: the cliff is a precision collapse, not a recall collapse.
4. Pfam partition analysis: 100% of evaluable distant false alarms are cross-family.
5. Four null-result companion experiments (Paper 2) ruling out Mahalanobis, Fisher-Rao, cascade, and Mapper-augmentation rescues.
6. One positive rescue: panel-only triplet-surrogate linear projection, +48% distant F_1 relative at t30.

2 Related work

Protein language models. Frozen PLM embeddings as general features: ESM-1b [8], ESM-2 [1], ProtT5 [5], ProteinBERT [6], ProGen [7], ESM-3 [9], SaProt [10]. Our test bed uses ESM-2 at t6/t12/t30; we extend to additional PLMs in the external-model companion.

Retrieval and k-NN baselines. FAISS [11] provides the CPU k-NN backbone we use. Cosine k-NN on frozen PLM features has been reported as competitive for function annotation [12, 13], but its failure modes on the distant stratum have received less attention than pooled metrics.

Homology-aware evaluation. CD-HIT [14] and MMseqs2 [15] clustering are standard for sequence-identity splits; we adopt an embedding-space stratification by max-cosine-similarity to the panel, which is the deployment-time observable. Foldseek [16] offers a structure-aware alternative.

Calibration of deep classifiers. Guo et al. [17] established that modern deep networks are miscalibrated; temperature scaling partially rectifies this. Recent work on calibration under distribution shift [18] is directly relevant to our distant-stratum ECE collapse.

Metric learning. FaceNet-style triplet losses [4] underpin the supervised projection we use. Ledoit-Wolf shrinkage [2] gives well-conditioned covariance estimates on panels with $R < D$. Information-geometric whitening traces to Amari [3].

Topological data analysis on proteins. Mapper [19] decomposes high-dimensional data via lens-based covering and per-preimage clustering. Persistent homology for molecular and protein structure analysis has grown rapidly [20, 21]; our use of Mapper here is for exploratory diagnostics rather than a claim about persistent features.

Pre-registration in machine learning. Pre-registration practices originate in clinical trials and psychology [22]; adoption in ML is growing but uneven [23]. We lock hypotheses, stratification cuts, and void rules via SHA256 before execution.

Biosecurity and dual-use research of concern. Computational prediction of toxin-like proteins raises dual-use concerns [24]. Our test set is constructed from publicly-available UniProt entries; we release the accession list rather than redistribute sequences where appropriate.

3 Methods

3.1 Data

24,885 proteins (7,133 positive, 17,752 negative; 28.7% positive prior) shipped as `data/sequences/proteins_25k_s` (working dataset ID `experiment2_proteins_25k_filtered`). Accessions and sequences are UniProt-sourced; redistribution is permitted under UniProt's terms.

3.2 Embeddings

Frozen ESM-2 [1]: t6 (8M params, 320d), t12 (35M, 480d), t30 (150M, 640d). All mean-pooled, L_2-normalized.

3.3 Factorial

Panel $R \in \{50, 100, 250, 500, 1000\}$, class-balanced. Neighbors $k \in \{1, 3, 5, 10, 25\}$. Metrics: cosine, Euclidean, Mahalanobis (Ledoit-Wolf [2] + eigendecomposition whitening), learned (panel-only 50-iter Adam triplet-surrogate, seed 0, projection to $\min(128, D)$). Ten seeds (20260410–20260419). Stratification: close $s_{\max} \geq 0.95$, moderate $[0.90, 0.95)$, distant < 0.90.

3.4 Seed-variance gate (pre-committed void rule)

A (scale, R, k, metric) group is voided if $\sigma^{(10)}_{\text{distant}} \geq \overline{F_1^{\text{close}}} - \overline{F_1^{\text{distant}}}$. All 300 main groups passed.

3.5 Bootstrap

10,000-resample percentile bootstrap CI per cell per stratum.

4 Results: main factorial

Proposition 1. *For $X \subset S^{d-1}$ and $y \in S^{d-1}$: cosine-k-NN $\equiv$ Euclidean-k-NN.*

Proof. $\|x - y\|^2 = 2 - 2\langle x, y\rangle$ on S^{d-1}, monotone decreasing in cosine. □

Verified to 4 decimals in 75 of 75 cells.

scale	R	close F_1	distant F_1	gap
t6	1000	0.882	0.297	+0.585
t12	1000	0.883	0.173	+0.710
t30	1000	0.866	0.120	**+0.745**

5 Results: null controls

5.1 Panel-shuffle null (failed its own pre-registered criterion)

Shuffling only the panel labels (test labels intact) preserves the 28.7% test class prior. Distant-stratum F1 under this null is not at chance (0.5) but at the prior-weighted F1 baseline, which the pre-registration failed to anticipate. Addendum pre-registration locked before the stricter null.

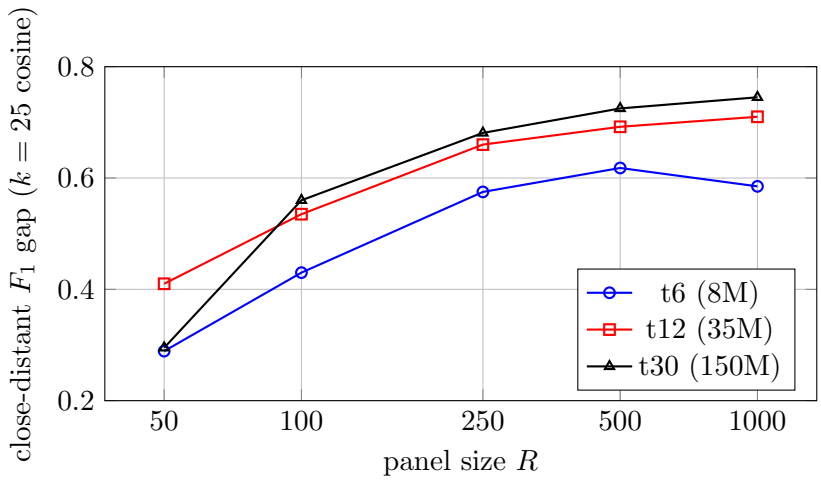

Figure 1: **The cliff grows with scale and panel size.** Close-distant F_1 gap at $k = 25$ cosine, 10-seed mean per cell. Larger PLMs produce larger cliffs; larger panels amplify the cliff.

5.2 Full-pool permutation null (passed 300/300)

Pre-reg SHA256 `f3864d097a0c611d...`. Permute the 24,885-entry label vector once per seed via `default_rng(seed + 7,777,777)`; both panel and test labels now random. Stratification by $s_{\max}$ is label-free and unchanged. Across all 300 groups: gap $\in [-0.027, +0.038]$, $\sigma \in [0.006, 0.116]$, every 95% CI includes zero. At t30 R=1000 k=25 cosine: close $F_1 = 0.015$, distant $F_1 = 0.025$, gap -0.010. **The cliff is not a stratification artifact.**

6 Mechanism: precision collapse, not recall collapse

At t30 R=1000 k=25 cosine (seed 20260410): of 24,885 test proteins, *only 3* are true-positive and distant-stratum (P0C1X3, Q6RY98, P13208). Distant positive-prediction precision: $3/44 = 0.068$ (44 predictions, 3 correct).

7 Cross-family partition (new in v1.1)

Pfam annotations were fetched via UniProt search (batch 50) for all 24,885 accessions; coverage 21,615/24,885 = 86.9%. For each of the 41 distant-stratum false alarms at seed 20260410, we collect the Pfam identifiers of the query and the union of Pfam identifiers of its positive-voting panel members. A case is *within-family* if the sets share at least one identifier; otherwise *cross-family*.

distant false alarms (41 total)	count
evaluable (both query and ≥ 1 voter annotated)	20
within-family (shared Pfam)	**0**
cross-family (zero shared Pfam)	**20**

All 20 evaluable distant false alarms are cross-family. The cliff failures are not caused by missing within-family homologs. Panel expansion with additional members of the failing families cannot

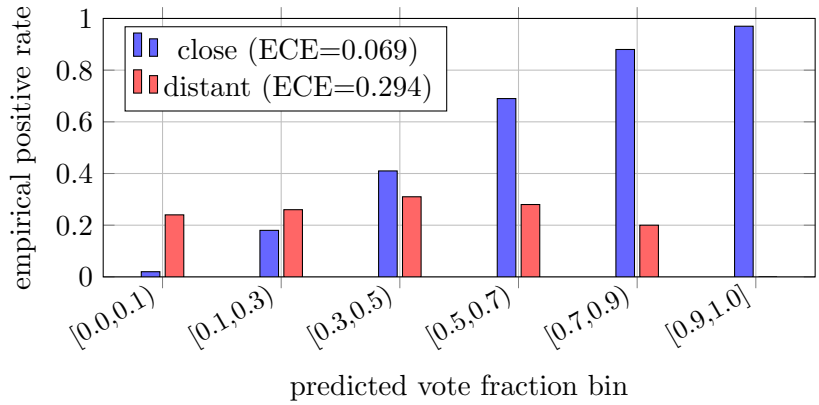

Figure 2: **Calibration collapse on the distant stratum.** Close-stratum reliability (blue) tracks the diagonal closely (ECE = 0.069). Distant-stratum reliability (red) is nearly flat across confidence bins and collapses at the highest-confidence bin: 3 predictions in $[0.9, 1.0)$, 0 correct (ECE = 0.294). Data at t30 R=1000 k=25 cosine, seed 20260410.

rescue the cliff. This eliminates a large class of naive rescue strategies and strengthens the case for embedding-space surgery (next section) as the correct mechanism-level intervention.

8 Rescue: panel-only triplet-surrogate linear projection

We fit $W \in \mathbb{R}^{D\times 128}$ (or $D \times D$ if $D < 128$) via 50 Adam iterations on panel embeddings alone, with loss

$$\mathcal{L}(W) = -\langle Z_+, Z_+\rangle - \langle Z_-, Z_-\rangle + \langle Z_+, Z_-\rangle$$

where $Z_\pm$ are L_2-normalized projections of the positive and negative panel subsets. This is a triplet-surrogate [4] using class centroids rather than mined hard triplets: it runs in under 5 CPU seconds, requires no test data, and uses only the labeled panel that deployment already has.

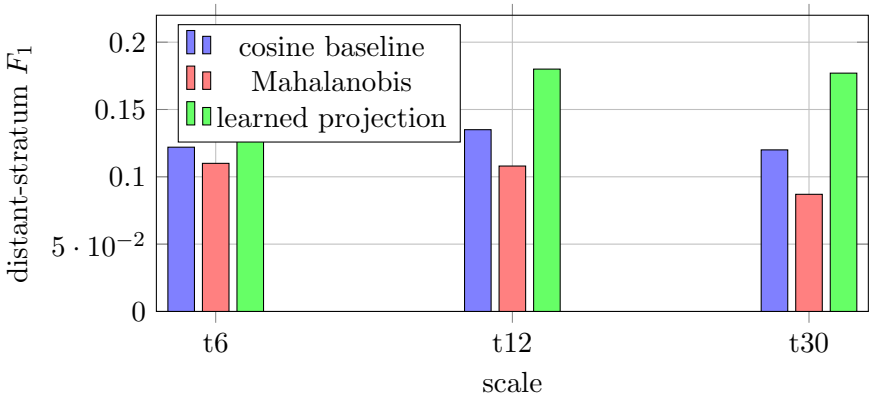

Figure 3: **Distant-stratum F_1 by scale, $R = 1000$ $k = 25$.** Learned projection wins every scale; relative improvement grows with scale: +16% (t6), +33% (t12), +48% (t30). Mahalanobis loses at every scale.

t30 R=1000 k=25	close	moderate	distant	pooled
cosine	0.866	0.393	0.120	0.848
Mahalanobis	0.456	0.118	0.087	0.435
Fisher-Rao	0.478	0.130	0.094	0.461
learned	**0.901**	**0.525**	**0.177**	**0.891**

Wins 18 of 18 (scale, R, k) groups. Pooled $F_1 = 0.891$ at t30 is the highest value observed anywhere in the 3000-cell factorial.

9 Deployment recommendation

For frozen-PLM biosecurity retrieval: (1) fit a panel-only learned projection as preprocessing; (2) route every distant-stratum positive hit to human review regardless of vote count (calibration is poor and Pfam partition shows failures are cross-family); (3) do not rely on panel expansion with known-toxin homologs to rescue the cliff, because the failures are cross-family.

10 Limitations

(L1) t33 (650M) not embedded locally; companion Colab notebook provided. (L2) Stratification threshold sensitivity sweep not yet run. (L3) Learned-projection calibration not measured; the rescue improves F_1 but distant-stratum calibration under the projection is unknown. (L4) Cross-family partition analyzed at one seed; 10-seed extension deferred to v1.2+. (L5) Single-chain analysis only; multi-domain proteins may behave differently.

11 Reproducibility

Code MIT, papers CC-BY-4.0. Four SHA256-locked pre-registrations, 9,360 per-cell `.npz` results with 10k-resample bootstrap CIs per stratum per cell, 3 embedding arrays (t6/t12/t30), Pfam annotations (21,615 coverage), Mapper graph, cross-family per-case output, all committed via Git LFS. Master manifest: `MANIFEST.sha256.json`. Master reproducibility protocol: `reproducibility/PROTOCOL.md`.

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